INCH-POUND

MIL-M-14/3
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, MELAMINE, THERMOSETTING, CONTAINING CELLULOSE, MINERAL, OR GLASS FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Melamine resin molding compounds, shall be of the following types, as specified:

Type CMG	–	Cellulose filler, general-purpose
Type CMI-5	–	Cellulose filler, impact-resistant; nominal impact strength, 0.5 foot-pound per inch notch
Type CMI-10	–	Cellulose filler, impact-resistant; nominal-impact strength, 1.0 foot-pound per inch notch
Type MME	–	Mineral filler, arc- and flame-resistant
Type MM1-5	–	Glass fiber filler, impact-resistant; nominal-impact strength, 0.5 foot-pounds per inch notch
Type MMI-30	–	Glass fiber filler, impact-resistant; nominal-impact strength, 3.0 foot-pounds per inch notch.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

AMSC N/A FSC 9330

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:55Z
Check the source to verify that this is the current version before use.

NOTES:

Type CMG. This type is a cellulose-filled, general-purpose, melamine molding compound.

Type CMI-5. This type is a cellulose-filled, moderate-impact, melamine compound.

Type CMI-10. This type is a cellulose-filled, moderate-impact, phenol modified melamine compound.

Type MME. This type is a mineral-filled melamine compound for use where good dielectric properties and arc and flame resistance are required.

Type MMI-5. This type is a glass fiber-filled melamine resin molding compound of lower-impact strength than type MMI-30, but with superior molding properties.

Type MMI-30. This type is a glass fiber-filled melamine compound of high-impact strength for use where heat resistance, arc resistance, and flame resistance are required.

TABLE I. *Qualification test requirements for melamine resin molding compounds, cellulose, mineral/glass filled.*

Requirement	Type CMG	Type CMI-5	Type CMI-10	Type MME	Type MMI-5	Type MMI-30
			Mechanical/physical			
Compressive strength endwise	20,000	25,000	–	25,000	28,000	20,000
Dimensional stability	–	–	–	0.7	–	
Flexural strength	8,000	7,500	11,000	6,000	8,500	8,500
Heat deflection temperature	120	160	–	120	200	200
Heat resistance	110	110	110	110	110	110
Impact strength, side[1]	–	0.5	1.0	–	0.5	3.0
Tensile strength	5,000	5,700	–	4,200	5,000	5,500
Water absorption	4.0	4.0	4.0	0.5	1.5	4.0
			Electrical			
Arc resistance	100	125	–	125	180	180
Dielectric breakdown:						
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	30	18	–	40	40	40
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	5	2.5	–	35	5	5

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:55Z
Check the source to verify that this is the current version before use.

TABLE I. *Qualification test requirements for melamine resin molding compounds, cellulose, mineral/glass filled – Continued.*

Requirement	Type CMG	Type CMI-5	Type CMI-10	Type MME	Type MMI-5	Type MMI-30
			Electrical *(Continued)*			
Dielectric constant:						
at 1 kilohertz	–	–	–	7.0	9.6	8.0
	–	–	–	7.0	10.0	9.0
at 1 megahertz	–	–	–	6.5	7.5	7.5
	–	–	–	6.5	8.0	8.0
Dielectric strength:						
Short-time test	275	250	–	325	215	150
Step-by-step test	200	150	–	275	150	125
Short-time test	125	50	–	275	100	100
Step-by-step test	100	30	–	225	50	50
Dissipation factor:						
at 1 kilohertz	–	–	–	0.06	0.08	0.06
	–	–	–	0.07	0.10	0.08
at 1 megahertz	–	–	–	0.04	0.03	0.03
	–	–	–	0.05	0.04	0.04
			Combustion			
Flame resistance:						
Ignition time	60	120	–	90	600	600
Burning time	180	60	–	180	0	0
Flammability	V-0	–	–	V-1	V-1	V-1
Toxicity when heated:						
Carbon dioxide	15,000	15,000	15,000	15,000	15,000	15,000
Carbon monoxide	1,000	1,000	1,000	1,000	1,000	1,000
Ammonia	2,500	2,500	2,500	2,500	2,500	2,500
Aldehydes as HCHO	50	50	50	50	50	50
Cyanide as HCN	60	60	60	60	60	60
Oxide of nitrogen at NO_2	100	100	100	100	100	100
Hydrogen chloride	100	100	100	100	100	100

[1] The side of the test specimen in that area formed by the chase of the mold.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:55Z
Check the source to verify that this is the current version before use.

TABLE II. *Batch acceptance test requirements for melamine resin molding compounds, cellulose, mineral or glass filled.*

Property to be tested	Type CMG	Type CMI-5	Type CMI-10	Type MME	Type MMI-5	Type MMI-30
	Performance requirement for each compound					
Arc resistance	100	125	–	125	180	180
Dielectric constant at 1 megahertz	–	–	–	6.5	8.0	8.0
Dielectric strength, step-by-step	100	30	–	225	50	50
Dissipation factor at 1 megahertz	–	–	–	0.05	0.04	0.04
Flexural strength	8,000	7,500	11,000	6,000	8,500	8,500
Impact strength, side[1]	–	0.5	1.0	–	0.5	3.0
Water absorption	4.0	4.0	4.0	0.5	1.5	4.0

[1]The side of the test specimen is that area formed by the chase of the mold.

Custodians:
Army – ME
Navy – SH
Air Force – 20

Review activities:
Army – ER, MI, MR, AR, EA
Navy – AS, OS
DLA – GS

User activities:
Navy – YD, EC

Preparing activity:
Navy - SH
(Project 9330-1097-04)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T15:55Z
Check the source to verify that this is the current version before use.